Develop and compare CLI, GUI, and Voice User Interfaces (VUI) for the same task and assess user satisfaction using Python (Tkinter for GUI, Speech Recognition for VUI), Terminal

Aim:

To develop and compare **Command Line Interface (CLI), Graphical User Interface (GUI), and Voice User Interface (VUI)** for a **Shopping Cart application** using Python (Terminal for CLI, Tkinter for GUI, and Speech Recognition for VUI) and evaluate user satisfaction.

PROCEDURE:

(I)CLI(Command Line Interface)

```
cart = []
def add_item(item):
    cart.append(item)
    print(f"{item} added to cart.")
def view_cart():
    if cart:
        print("\nItems in your cart:")
        for i, item in enumerate(cart, 1):
            print(f"{i}. {item}")
    else:
        print("Cart is empty.")
def remove_item(index):
    if 0 < index <= len(cart):
        removed = cart.pop(index-1)
        print(f"{removed} removed from cart.")
    else:
        print("Invalid item number.")
```

```
def main():
    while True:
        print("\n1. Add Item  2. View Cart  3. Remove Item  4. Exit")
        choice = input("Enter choice: ")
        if choice == '1':
            item = input("Enter item name: ")
            add_item(item)
        elif choice == '2':
            view_cart()
        elif choice == '3':
            view_cart()
            num = int(input("Enter item number to remove: "))
            remove_item(num)
        elif choice == '4':
            print("Exiting...")
            break
        else:
            print("Invalid choice")
if __name__ == "__main__":
    main()
```

SAMPLE OUTPUT:

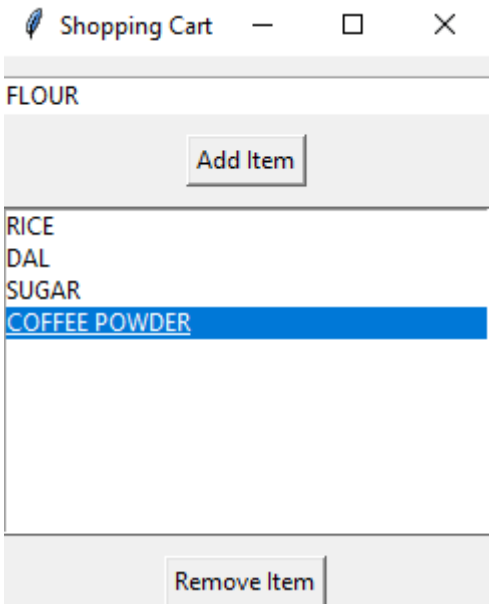
```
1. Add Item  2. View Cart  3. Remove Item  4. Exit
Enter choice: 1
Enter item name: RICE
RICE added to cart.

1. Add Item  2. View Cart  3. Remove Item  4. Exit
Enter choice: 1
Enter item name: DAL
DAL added to cart.

1. Add Item  2. View Cart  3. Remove Item  4. Exit
Enter choice: 2

Items in your cart:
1. RICE
2. DAL

1. Add Item  2. View Cart  3. Remove Item  4. Exit
Enter choice: 3

Items in your cart:
1. RICE
2. DAL
Enter item number to remove: 2
DAL removed from cart.

1. Add Item  2. View Cart  3. Remove Item  4. Exit
Enter choice: 4
Exiting...
```

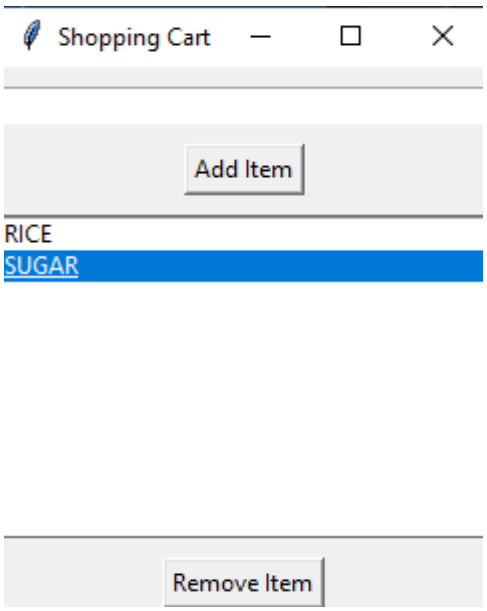
(II)GUI(Graphical User Interface)

```
import tkinter as tk

from tkinter import messagebox

cart = []

def add_item():

    item = item_entry.get()

    if item:

        cart.append(item)

        item_entry.delete(0, tk.END)

        update_list()

    else:

        messagebox.showwarning("Warning","Item cannot be empty")

def update_list():

    cart_list.delete(0, tk.END)

    for item in cart:

        cart_list.insert(tk.END, item)
```

```
def remove_item():
    selected = cart_list.curselection()
    if selected:
        cart.pop(selected[0])
        update_list()
app = tk.Tk()
app.title("Shopping Cart")
item_entry = tk.Entry(app,width=40)
item_entry.pack(pady=10)
add_btn = tk.Button(app,text="Add Item",command=add_item)
add_btn.pack()
cart_list = tk.Listbox(app,width=40)
cart_list.pack(pady=10)
remove_btn = tk.Button(app,text="Remove Item",command=remove_item)
remove_btn.pack()
app.mainloop()
```

SAMPLE OUTPUT:





(III)VLI(Voice User Interface)

```
import speech_recognition as sr
```

```
cart = []
r = sr.Recognizer()
def listen():
    with sr.Microphone() as source:
        print("Speak command...")
        audio = r.listen(source)

    try:
        command = r.recognize_google(audio)
        print("You said:", command)
        return command.lower()
    except:
        return ""
while True:
    cmd = listen()
    if "add" in cmd:
        item = cmd.replace("add","").strip()
        cart.append(item)
        print(item,"added")
    elif "show cart" in cmd:
        print("Cart:",cart)
    elif "remove" in cmd:
        item = cmd.replace("remove","").strip()
        if item in cart:
            cart.remove(item)
            print(item,"removed")
    elif "exit" in cmd:
        break
```

SAMPLE OUTPUT:

```
Speak command...
You said: add eggs
eggs added

Speak command...
You said: add chocolate
chocolate added

Speak command...
You said: show cart
Cart: ['eggs', 'chocolate']

Speak command...
You said: remove eggs
eggs removed

Speak command...
You said: exit
```